

International Information & Library Review

ISSN: 1057-2317 (Print) 1095-9297 (Online) Journal homepage: www.tandfonline.com/journals/ulbr20

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**To cite this article:** M. P. Singh & Ankit Kumar (30 Aug 2026): Digital Literacy, ChatGPT Adoption and AI Integration Among LIS Research Scholars in Selected Central Universities in North India, International Information & Library Review, DOI: 10.1080/10572317.2026.2722824

**To link to this article:** https://doi.org/10.1080/10572317.2026.2722824

Published online: 30 Aug 2026.

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INTERNATIONAL INFORMATION & LIBRARY REVIEW
https://doi.org/10.1080/10572317.2026.2722824

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# Digital Literacy, ChatGPT Adoption and AI Integration Among LIS Research Scholars in Selected Central Universities in North India

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**ABSTRACT**

The integration of Artificial Intelligence (AI) tools such as ChatGPT is rapidly transforming academic and research practices in higher education by influencing how research scholars access, evaluate and generate information. Within the field of Library and Information Science (LIS), the increasing adoption of ChatGPT has created new opportunities for research support, scholarly productivity and AI integration while simultaneously raising concerns regarding academic integrity, ethical AI use, institutional preparedness and responsible adoption. However, empirical evidence on digital literacy, ChatGPT adoption, and AI integration among LIS research scholars in Indian higher education remains limited. This study aims to examine digital literacy, ChatGPT adoption, and AI integration among LIS research scholars in selected Central Universities in North India. Specifically, the study applies the Unified Theory of Acceptance and Use of Technology (UTAUT) to investigate the factors influencing research scholars' perceptions and adoption of ChatGPT in academic and research activities. A quantitative, descriptive and cross-sectional research design was adopted. Data were collected using a structured questionnaire comprising both closed- and open-ended questions. During the (2024–2025) academic year 212 valid responses were obtained from LIS research scholars enrolled in five selected Central Universities in North India. Descriptive statistical techniques and Welch's t-test were employed to analyze respondents' perceptions and examine gender-based differences. The findings indicate that LIS research scholars demonstrate a generally positive perception of ChatGPT, particularly with respect to Performance Expectancy (PE), Effort Expectancy (EE) and Social Influence (SI). Significant gender-based differences were observed in Performance Expectancy. Respondents acknowledged ChatGPT's benefits in improving research efficiency, enhancing information accessibility, supporting academic writing, and facilitating AI integration in research activities. Nevertheless, concerns regarding overreliance on AI, limited institutional AI policies, inadequate faculty guidance, and ethical challenges associated with AI use were also identified. The findings further highlight that digital literacy plays a critical role in supporting the responsible adoption of ChatGPT and AI technologies in higher education. Unlike previous studies that primarily examine general AI adoption, this study extends the application of the UTAUT framework by integrating digital literacy, ChatGPT adoption, and AI integration within the context of LIS research scholars in selected Central Universities in North India. The study provides empirical evidence that contributes to the growing body of knowledge on AI-enabled research practices and offers practical implications for universities, librarians, faculty members, and policymakers to strengthen institutional AI policies, enhance digital literacy programmes, and promote the responsible integration of AI technologies in academic and research environments.

**KEYWORDS**

Academic Integrity; Digital Literacy; ChatGPT; UTAUT Model; AI Tools; Artificial Intelligence (AI)

## Introduction

Artificial Intelligence (AI) is increasingly transforming teaching, learning, and research practices in higher education by reshaping the way knowledge is accessed, generated, and disseminated. Among recent AI innovations, ChatGPT, developed by OpenAI in 2022, has attracted considerable attention owing to its ability to generate human-like responses, retrieve and synthesize information, support academic writing, and assist researchers in various scholarly activities. The rapid adoption of generative AI technologies has created new opportunities for improving research productivity, accelerating information

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

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access, and enhancing academic collaboration. Consequently, AI-powered applications are becoming an integral part of contemporary research environments.

In the field of Library and Information Science (LIS), AI-driven tools such as ChatGPT have considerable potential to improve information retrieval, digital reference services, metadata generation, literature searching, scholarly communication, and research support services. However, the growing integration of AI into academic and research environments has also raised important concerns regarding digital literacy, ethical AI use, academic integrity, misinformation, data privacy, plagiarism, algorithmic bias, and excessive dependence on AI-generated content. As generative AI increasingly influences scholarly communication and knowledge production, it has become essential to understand how research scholars perceive, evaluate, and responsibly adopt these technologies within academic settings.

Although previous studies have examined AI adoption, technology acceptance, and ChatGPT usage in higher education, the majority have focused on general student populations, perceived usefulness, learning outcomes, or instructional applications. Limited empirical evidence is available on digital literacy, ChatGPT adoption, and AI integration among Library and Information Science research scholars in Indian Central Universities. Moreover, existing studies have rarely examined these issues from the perspective of the Unified Theory of Acceptance and Use of Technology (UTAUT) while simultaneously considering digital literacy, ethical concerns, institutional preparedness, and responsible AI adoption. Consequently, the theoretical understanding of AI adoption among LIS research scholars remains limited.

To address this research gap, the present study investigates digital literacy, ChatGPT adoption, and AI integration among LIS research scholars at the University of Delhi (DU), Aligarh Muslim University (AMU), Banaras Hindu University (BHU), Central University of Punjab (CUPB), and Babasaheb Bhimrao Ambedkar University (BBAU). The study applies the Unified Theory of Acceptance and Use of Technology (UTAUT) to examine the influence of Performance Expectancy (PE), Effort Expectancy (EE), and Social Influence (SI) on scholars' perceptions and adoption of ChatGPT in academic and research activities. By integrating digital literacy and responsible AI use within the UTAUT framework, this study extends the application of the model beyond its routine use in technology acceptance research and provides empirical evidence from the Indian LIS context. The findings are expected to contribute to the growing literature on AI-enabled research practices while offering practical implications for universities, librarians, faculty members, and policymakers to strengthen digital literacy initiatives, formulate institutional AI policies, and promote the ethical and responsible integration of AI technologies in higher education.

## Review of related literature

### *AI adoption and academic use of ChatGPT*

Since the launch of ChatGPT by OpenAI in 2022, the integration of Artificial Intelligence (AI) in higher education has gained considerable scholarly attention. Studies indicate that ChatGPT is increasingly used for content generation, idea development, literature review, summarization, language enhancement, and academic writing support. Tlili et al. (2023) found that public perception toward ChatGPT in education is generally positive, particularly regarding its usefulness in improving learning efficiency and accessibility. Similarly, Kasneci et al. (2023) highlighted that ChatGPT facilitates collaborative learning, personalized instruction, and research assistance, while Mogavi et al. (2023) reported that both students and educators perceive ChatGPT as a transformative educational technology that enhances learner motivation and engagement. Collectively, these studies demonstrate the growing educational potential of generative AI in higher education.

However, recent studies suggest that the successful adoption of ChatGPT depends not only on its technological capabilities but also on users' behavioral intentions, digital competencies, and institutional support. Subhani et al. (2025), using the UTAUT framework, reported that

performance expectancy, effort expectancy, social influence, self-efficacy, hedonic motivation, and price value significantly influence students' intention to adopt ChatGPT, while academic integrity positively moderates the relationship between adoption intention and actual use. These findings indicate that responsible AI adoption extends beyond technological usefulness and requires appropriate ethical awareness and user preparedness.

Recent studies from developing countries further emphasize the importance of institutional readiness in AI adoption. Hamamra et al. (2026) reported that Palestinian educators recognized AI's potential to improve educational quality, although concerns regarding institutional preparedness and teaching practices remained significant. In the Indian context, Sharma and Singh (2024) found that technological readiness and user acceptance significantly influence AI adoption in universities. Similarly, Kumar et al. revealed that Indian students demonstrate positive perceptions and readiness toward AI-enhanced learning environments. From the perspective of Library and Information Science (LIS), Aithal and Aithal (2023) argued that ChatGPT can complement traditional library services by improving information retrieval, digital reference services, and personalized user support. Although existing studies consistently report the educational benefits of ChatGPT, comparatively limited attention has been given to its adoption among LIS research scholars, particularly within Indian Central Universities.

### *Ethical concerns and academic integrity*

Despite its educational advantages, several researchers have highlighted ethical concerns associated with ChatGPT usage in academia. Cotton et al. (2024) and Vaishya et al. (2023) identified plagiarism, misinformation, privacy risks, and academic dishonesty as major challenges associated with AI-generated content. Similarly, Susnjak and McIntosh (2024) argued that ChatGPT's ability to generate human-like academic responses threatens the integrity of online assessments and scholarly writing. Supporting this argument, Van Wyk, (2023) found that AI-generated content is increasingly difficult to distinguish from human-written work, raising concerns regarding originality, transparency, and accountability in academic research.

Alhur et al. reported that excessive dependence on AI tools may reduce critical thinking, creativity, and academic autonomy among educators and learners. Likewise, Khlaif et al. (2025) identified inadequate institutional AI policies, limited faculty training, insufficient digital competencies, and concerns regarding the quality of AI-generated outputs as major barriers to responsible AI integration. Supporting these findings, Shahzad et al. (2025) emphasized that successful implementation of generative AI requires institutional readiness, technological infrastructure, teacher support, and clearly defined ethical guidelines. These findings collectively suggest that responsible AI integration requires institutional governance, ethical awareness, and balanced use of AI technologies in higher education.

### *UTAUT model and AI adoption*

The Unified Theory of Acceptance and Use of Technology (UTAUT), proposed by Venkatesh et al. (2003), is one of the most widely accepted theoretical frameworks for explaining technology adoption behavior. The model comprises four major constructs Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), and Facilitating Conditions (FC) which significantly influence users' behavioral intention to adopt new technologies. Owing to its strong predictive capability, the UTAUT framework has been extensively applied in studies examining AI adoption in higher education.

Ayyoub et al. (2025) reported that performance expectancy, effort expectancy, and social influence significantly influence educators' behavioral intention to use generative AI tools. Similarly, Lee et al. (2024) found that perceived usefulness and ease of use positively affect ChatGPT adoption. In the Indian context, Sharma and Singh (2024) confirmed that institutional support and technological readiness significantly influence AI acceptance in universities. Furthermore, Shahzad et al. (2025) demonstrated that digital literacy plays a significant role in shaping behavioral

intention toward generative AI through attitude, subjective norms, and perceived behavioral control.

Although previous studies have successfully applied UTAUT and related theoretical models to explain AI adoption, they have primarily focused on general student populations and educators. Limited attention has been given to Library and Information Science (LIS) research scholars, whose academic activities require advanced digital literacy, responsible AI use, and ethical information practices. Therefore, the present study extends the application of the UTAUT framework by examining ChatGPT adoption in relation to digital literacy and AI integration among LIS research scholars in selected Central Universities of North India, thereby contributing to the growing literature on responsible AI adoption in higher education.

### *Research gap*

Existing literature demonstrates increasing scholarly interest in ChatGPT adoption, generative AI, digital literacy, and technology acceptance in higher education. However, most previous studies have primarily focused on students, educators, or general university populations, while comparatively limited attention has been given to Library and Information Science (LIS) research scholars, particularly within the Indian higher education context. Furthermore, earlier investigations have largely emphasized the functional benefits of AI tools, with relatively limited discussion on the combined role of digital literacy, ethical AI use, institutional preparedness, and responsible AI adoption.

In addition, although recent studies have applied UTAUT framework to explain AI adoption, very few have examined ChatGPT adoption among LIS research scholars across multiple Central Universities in India. Therefore, the present study addresses this theoretical and empirical gap by investigating digital literacy, ChatGPT adoption, AI integration, ethical concerns, and behavioral perceptions among LIS research scholars in five selected Central Universities of North India using the UTAUT framework. The study also extends the application of UTAUT by examining responsible AI adoption within the context of scholarly research, thereby providing empirical evidence that may assist universities, librarians, and policymakers in developing effective strategies for the responsible integration of generative AI in higher education.

## Aim and objectives for this study

The aim and objectives of this study are as follows:

1. To assess the level of awareness and digital literacy regarding ChatGPT among LIS research scholars.
2. To explore the purpose and frequency of ChatGPT usage in academic and research activities.
3. To examine perceptions regarding the accuracy, reliability, and usefulness of ChatGPT.
4. To examine the ethical concerns, academic integrity issues and responsible use of ChatGPT in higher education.
5. To provide recommendations for promoting the responsible integration of AI tools in academic and research environments.

## Research methodology

### *Research design*

This study adopted a quantitative research design with a descriptive and cross-sectional survey approach to examine digital literacy, ChatGPT adoption, and AI integration among LIS research scholars. The quantitative approach was considered appropriate as the study aimed to measure respondents' perceptions numerically using selected constructs of the Unified Theory of Acceptance and Use of Technology (UTAUT), namely Performance Expectancy (PE), Effort Expectancy (EE), and Social Influence (SI).

### *Population of the study*

The target population comprised research scholars from the Departments of Library and Information Science (DLIS) of five selected

Central Universities in India during the academic year 2024–25. The total population of the study was 260 research scholars ($N = 260$).

### Sample size determination

The sample size was calculated using the finite population correction formula at a 95% confidence level and 5% margin of error, resulting in a required minimum sample size of approximately 156 respondents. However, the study successfully collected 212 valid responses, which exceeded the required sample size. The achieved sample exceeded the minimum required sample size, thereby enhancing the statistical reliability and generalizability of the study findings.

### Sampling technique

A non-probability convenience sampling technique was used for selecting respondents. This method was adopted due to the accessibility of research scholars across multiple universities and the practical limitations of reaching the entire population. Only those respondents who were available and willing to participate were included in the study.

### Data collection method

Primary data were collected using a structured questionnaire administered online. The questionnaire was distributed through WhatsApp and email platforms to ensure wider accessibility and quicker response collection. The data collection period was from 17 April to 30 June 2025. Prior informed consent was obtained electronically from all respondents before participation in the survey. This duration allowed for repeated reminders and ensured a higher response rate.

To ensure ethical compliance:

- Participation was voluntary
- Respondents were informed about the purpose of the study
- No personal identifiers were collected
- Data were used strictly for academic research purposes
- Duplicate responses were minimized using controlled online form settings (single-response restriction)

### Research instrument

The structured questionnaire was developed after an extensive review of the literature on ChatGPT adoption, digital literacy, and the Unified Theory of Acceptance and Use of Technology (UTAUT). The questionnaire was designed by the researcher in accordance with the objectives of the study and guided by the theoretical concepts of the UTAUT framework proposed by Venkatesh et al. (2003). The instrument included items measuring Performance Expectancy (PE), Effort Expectancy (EE), Social Influence (SI), awareness, usage behavior, perceived advantages, limitations, and ethical concerns related to ChatGPT in academic and research activities.

### Validity and reliability

To ensure content validity, the questionnaire was evaluated by subject experts in the field of Library and Information Science and research methodology. Their suggestions regarding the clarity, relevance, and wording of the items were incorporated before finalizing the instrument. A pilot study was conducted with a small group of research scholars to assess the clarity, structure, and comprehensibility of the questionnaire, and minor modifications were made based on the feedback received. The internal consistency of the instrument was assessed using Cronbach's Alpha, and all constructs recorded reliability coefficients above the recommended threshold of 0.70, indicating satisfactory reliability of the instrument. Since the questionnaire was developed on the basis of an extensive review of the literature, guided by the UTAUT framework, and reviewed by subject experts, adequate content validity of the instrument was established.

### Hypotheses of the study

The following hypotheses were formulated based on selected UTAUT constructs (PE, EE, and SI):

#### *Null Hypothesis ($H_0$):*

There is no significant difference between research scholars' mean scores on UTAUT constructs (Performance Expectancy, Effort Expectancy, and Social Influence) and the neutral value, indicating no strong acceptance or rejection of ChatGPT for academic use.

#### *Alternative Hypothesis ($H_1$):*

There is a significant difference between research scholars' mean scores on UTAUT constructs (Performance Expectancy, Effort Expectancy, and Social Influence) and the neutral value, indicating a measurable positive or negative perception toward the adoption of ChatGPT for academic use.

### *Data analysis*

The collected data were analyzed using IBM SPSS Statistics and Microsoft Excel. Descriptive statistics, including frequencies, percentages, means, and standard deviations, were used to summarize respondents' demographic characteristics and perceptions. Inferential statistical techniques, including the one-sample *t*-test and Welch's independent samples *t*-test, were employed to examine respondents' perceptions and gender-based differences. Statistical significance was determined at the 5% level ($p < 0.05$). The analysis was guided by the selected UTAUT constructs (Performance Expectancy, Effort Expectancy, and Social Influence) to understand ChatGPT adoption among LIS research scholars.

## Scope and limitations of the study

The scope of this study is limited to examining the use of ChatGPT among research scholars from the Department of Library and Information Science (DLIS) of selected Central Universities in North India, namely the University of Delhi (DU), Aligarh Muslim University (AMU), Banaras Hindu University (BHU), Central University of Punjab Bhatinda (CUPB) and Babasaheb Bhimrao Ambedkar University (BBAU). The study specifically focuses on three constructs of the UTAUT model Performance Expectancy (PE), Effort Expectancy (EE) and Social Influence (SI) to understand users' perceptions, awareness, and adoption behavior regarding ChatGPT in academic and research activities. The findings are based on primary data collected from 212 valid responses out of a population of 260 research scholars during the academic year 2024–25, making the study relevant to AI adoption in higher education and library science research environments.

However, the study has certain limitations. It is geographically restricted to selected Central Universities and DLIS departments only, which may limit the generalizability of the findings to other disciplines or institutions. The use of self-reported data may introduce response bias, as participants may not always provide fully accurate responses. Data collection was conducted through online platforms such as WhatsApp and Google Forms, which may have limitations related to accessibility, respondent engagement, and lack of direct control over the survey environment. The relatively short data collection period (17 April to 30 June 2025) may also restrict deeper insights into long-term usage patterns. Furthermore, the study adopts a cross-sectional design, capturing perceptions at a single point in time, and does not account for changes in user behavior over time, which a longitudinal study could better explore.

## Results of the study

The responses from 212 participants to 15 questions were organized into six sections: (i) Demographic information, (ii) Awareness and use of ChatGPT, (iii) Purpose of using ChatGPT, (iv) Advantages of ChatGPT, (v) Challenges of using ChatGPT, and (vi) Integrity and ethical considerations. The data was presented in tables and figures and analyzed using simple calculation methods.

### *Demographic information*

The demographic data, including education level, respondents' age, and gender, are major phenomena that lead to any research's core area (Table 1).

Table 1 indicates Out of the total 212 respondents, a majority were male 124 (58.5%), while

**Table 1.** Demographic information of respondents ($n = 212$).

| Demographic variable | Category | Frequency (n) | Percentage (%) |
|---|---|---|---|
| Gender | Male | 124 | 58.5% |
| | Female | 88 | 41.5% |
| Age group | 21–25 | 11 | 5.2% |
| | 26–30 | 133 | 62.7% |
| | Above 30 | 68 | 32.1% |
| University enrolled | University of Delhi (DU), Delhi | 52 | 24.5% |
| | Aligarh Muslim University (AMU), Aligarh | 38 | 18.0% |
| | Banaras Hindu University (BHU), Varanasi | 44 | 20.7% |
| | Central University of Punjab Bhatinda (CUPB), Punjab | 48 | 22.6% |
| | Babasaheb Bhimrao Ambedkar University (BBAU), Lucknow | 30 | 14.2% |

**Table 2.** Awareness and use of ChatGPT among LIS research scholars ($n = 212$).

| University name | Total population | No. of respondent | Aware of ChatGPT | Awareness (%) |
|---|---|---|---|---|
| University of Delhi (DU), Delhi | 64 | 52 | 52 | 100% awareness |
| Aligarh Muslim University (AMU), Aligarh | 50 | 38 | 38 | |
| Banaras Hindu University (BHU), Varanasi | 50 | 44 | 44 | |
| Central University of Punjab Bhatinda (CUPB), Punjab | 60 | 48 | 48 | |
| Babasaheb Bhimrao Ambedkar University (BBAU), Lucknow | 36 | 30 | 30 | |
| Total | 260 | 212 | 212 | 100% |

female respondents accounted for 88 (41.5%). This indicates a slightly higher male representation in the sample population. With regard to age, the largest proportion of participants 133 (62.7%) fell within the 26–30 years age group, only a small percentage are in the (21–25) respondents accounted for 11 (5.2%) and Respondents aged above 30 years comprised 68 (32.1%) of the sample. In terms of academic affiliation, respondent's despite of equal distribution (60 questionnaires per institution), the response rate varied slightly. DU contributed (24.5%) of the total responses ($n = 52$), while CUPB accounted for 22.6% ($n = 48$), BHU contributed 44 (20.7%), AMU contributed 38 (18%) responses and least contributed by BBAU 30 (14.2%). This near-equal participation ensures balanced representation from all five universities, contributing to the credibility and comparative strength of the data collected.

### *Awareness and use of ChatGPT*

"ChatGPT, a chatbot and virtual assistant developed by Open AI, was publicly released on November 30, 2022. This large language model, built on Open AI's generative pre-trained transformer (GPT) architecture, rapidly gained popularity in academic circles". Its broad array of features has drawn interest from various academic stakeholders, including students, scholars, faculty, educators, and instructors, eager to utilize its capabilities. However, ChatGPT also has its limitations table 2.

In Table 2, highlight that total of 212 research scholars participated in the study, including 52 (24.5%) from University of Delhi (DU), 48 (22.6%) from Central University of Punjab Bhatinda (CUPB), 44 (20.7%) from Banaras Hindu University (BHU), 38 (18%) from Aligarh Muslim University (AMU) and 30 (14.2%) from Babasaheb Bhimrao Ambedkar University (BBAU). All respondents 212 (100%) reported being aware of ChatGPT, indicating its widespread recognition among academic researchers.

### *Purpose of using ChatGPT*

Recent studies have highlighted concerns regarding the misuse of ChatGPT for academic dishonesty. The primary issue centers around the potential compromise of academic integrity, as students may rely on AI-generated content (particularly essays) that do not genuinely reflect their own understanding or efforts. This raises ethical concerns within academic institutions. In this context, participants were asked about their use of ChatGPT for academic purposes and how it supports their academic tasks. The findings are presented in Table 3.

In Table 3, reveals that a significant proportion of respondents are actively engaging with ChatGPT for academic tasks. Nearly half of the researchers' 95 (44.8%) reported using the tool

**Table 3.** Purpose of using ChatGPT ($n = 212$).

| S.No. | Particular | No. of Respondents (%) |
|---|---|---|
| | *Frequency of using ChatGPT* | |
| 1. | Daily | 63 (29.7%) |
| 2. | Weekly | 95 (44.8%) |
| 3. | Monthly | 42 (19.8%) |
| 4. | Rarely | 12 (05.7%) |
| | *Types of academic tasks do you use for** | |
| 1. | Generating ideas for assignments/projects | 171 (80.7%) |
| 2. | Text and content generation | 140 (66.0%) |
| 3. | Designing powerpoint presentation and graphic works | 85 (40.1%) |
| 4. | Paraphrase the content | 58 (27.3%) |
| 5. | Spellings and grammar correction | 61 (28.8%) |
| 6. | Review of related literature | 89 (42.0%) |
| 7. | Analysis and summarize the article | 130 (61.3%) |
| | *Perceive the accuracy of information provided by ChatGPT* | |
| 1. | Somewhat accurate | 99 (46.7%) |
| 2. | Not very accurate | 58 (27.4%) |
| 3. | Very accurate | 48 (22.6%) |
| 4. | Not accurate at all | 07 (03.3%) |
| | *Would you recommend using ChatGPT to other students for academic purposes?* | |
| 1. | Yes, highly likely | 85 (40.1%) |
| 2. | Maybe, somewhat likely | 120 (56.6%) |
| 3. | No, not likely | 07 (03.3%) |

Note
*Multiple answers are permitted.

on a weekly basis, followed by 63 (29.7%) who use it daily. This pattern indicates a growing reliance on ChatGPT for regular academic assistance.

When it comes to academic applications, a substantial number of scholars 171 (80.7%) use ChatGPT for generating ideas for assignments or projects, suggesting that the tool is primarily leveraged for brainstorming and content planning. Other commonly reported uses include text and content generation 140 (66%) and summarizing or analyzing articles, 130 (61.3%), showing its role in processing and simplifying complex academic material. Additionally, 89 (42%) of students use it for reviewing related literature, which highlights its contribution in the research phase of academic work. Less frequent uses include tasks like paraphrasing content 58 (27.3%) and correcting grammar and spelling 61 (28.8%), which indicates these are supplementary benefits rather than primary purposes.

Regarding the perceived accuracy of information provided by ChatGPT, a majority 99 (46.7%) found the tool to be somewhat accurate, while 48 (22.6%) considered it very accurate. However, a notable 58 (27.4%) expressed concerns about the tool's reliability by rating it as not very accurate. A small portion 07 (3.3%) found the information not accurate at all, suggesting that while the tool is generally trusted, users still approach its content with some caution. Similarly, when asked about recommending ChatGPT to peers, most of 120 (56.6%) responded that they are somewhat likely to recommend it, while 85 (40.1%) stated they are highly likely to do so. Only a small fraction 07 (3.3%) was unlikely to recommend it. This indicates an overall positive perception and satisfaction among users, despite minor reservations regarding accuracy.

### Advantages of using ChatGPT

Adopting ChatGPT for academic tasks offers considerable advantages to research scholars. It serves as a valuable tool for generating innovative ideas, refining arguments, and providing constructive feedback on critical concepts. The various benefits that LIS research scholars derive from ChatGPT are summarized in Figure 1.

Figure 1, presents the average responses of research scholars regarding the perceived benefits of ChatGPT in academic and research contexts. The highest-rated statements indicate strong positive perceptions of ChatGPT's utility in saving time and effort ($M = 4.57$) and reducing review and publication costs ($M = 4.30$). These findings reflect those respondents widely recognize ChatGPT as a valuable tool for streamlining research workflows and alleviating publication burdens. Moreover, participants acknowledged the platform's role in easing access to

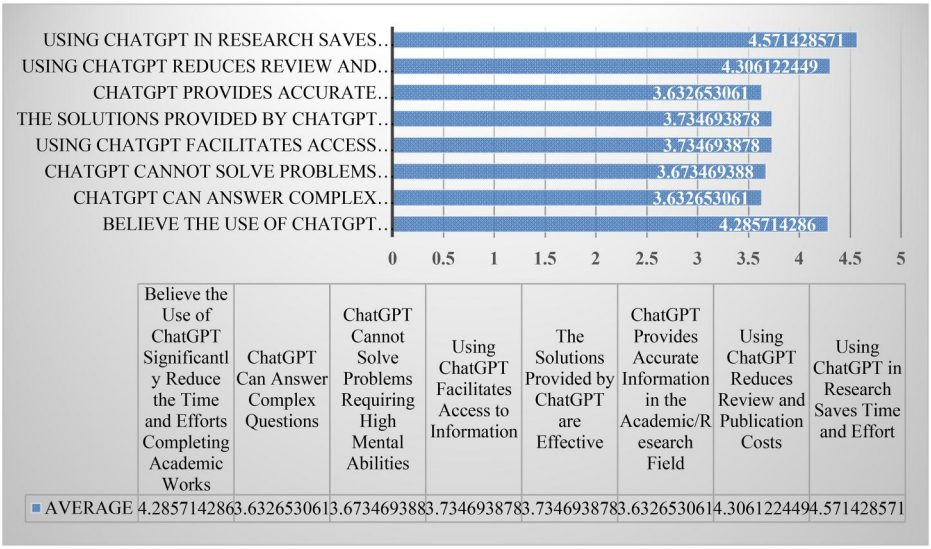



**Figure 1.** Conceptual framework for ChatGPT adoption among LIS research scholars. *Source: Authors' Own Work.

information ($M = 3.73$) and providing effective solutions ($M = 3.73$), suggesting moderate confidence in its problem-solving capabilities. The average score for ChatGPT's ability to answer complex questions ($M = 3.63$) and provide accurate academic information ($M = 3.63$) demonstrates a reasonable level of trust in its knowledge base, albeit with some caution. Interestingly, the statement that ChatGPT "cannot solve problems requiring high mental abilities" also received a moderate mean score ($M = 3.67$), indicating ambivalence among respondents some view ChatGPT as limited in handling tasks that require deep critical thinking or creativity.

### *Challenges of using ChatGPT*

The integration of ChatGPT presents several challenges for educators. Ensuring the accuracy and quality of the content generated by ChatGPT can be a struggle for educators. Monitoring students' use of ChatGPT to prevent misuse demands additional time, diverting their attention from other teaching responsibilities. Educators will have difficulty in assessing genuine learning achievements of students because it will be difficult to identify whether content is originally thought or ChatGPT generated (Grassini, 2023). There's a risk of educators facing skill obsolescence and becoming heavily reliant on ChatGPT for academic and administrative tasks (Figure 2).

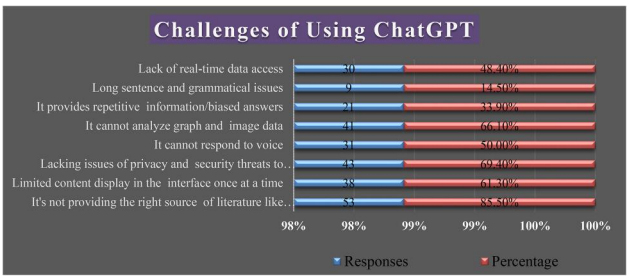



**Figure 2.** Challenges of using ChatGPT ($n = 212$).
**Note:** *Multiple answers are permitted.

The findings of the study reveal that several significant challenges faced by research scholars in using ChatGPT for academic purposes. The most commonly reported issue, identified by 53 (85.5%) of respondents, is its inability to provide reliable sources of literature such as books and academic articles, which limits its usefulness for scholarly writing. Privacy and data security concerns were raised by 43 (69.4%) of participants, reflecting hesitation in sharing sensitive information with the AI. Furthermore, 41 (66.1%) pointed out that ChatGPT cannot analyze graphs or image data, restricting its capacity for visual or data-driven research. Limited content display within the interface was also a concern for 38 (61.3%) of users, affecting the efficiency of longer research tasks. Additionally, 31 (50%) of respondents noted the lack of voice command functionality, and 30 (48.4%) reported dissatisfaction with ChatGPT's inability to access real-time data.

Other issues included repetitive or biased responses 21 (33.9%) and grammatical problems in lengthy outputs 09 (14.5%). Overall, while ChatGPT is a helpful academic tool, these challenges highlight important limitations in its current form that must be addressed to ensure more effective and ethical use in higher education and research.

### *Integrity and ethical consideration of using ChatGPT*

The integration of AI tools like ChatGPT into scholarly activities necessitates clear institutional guidelines to uphold academic integrity. Challenges arise in ensuring the responsible use of AI, especially in the absence of well-defined policies that address issues such as plagiarism, authorship, and transparency. Researchers emphasize the need for ethical awareness and structured protocols to navigate the complexities of AI-assisted academic work. The summarized responses reflecting these concerns are presented in Table 4.

Table 4, indicates a substantial ethical deficiency regarding the utilization of ChatGPT in academic institutions. Only 65 (30.7%) of research scholars respond that presence of institutional policies guiding its use, while 147 (69.3%) indicated absence of any such framework. This lack of formal regulation leads to several ethical concerns from a researcher's perspective. 1) Ambiguity regarding authorship and intellectual ownership, as content generated through ChatGPT may compromise the authenticity of scholarly work. 2) Improper citation of ChatGPT responses raises questions about academic transparency. 3) Data privacy risks emerge when sensitive research queries or unpublished information are input into the tool. 4) Potential for plagiarism or academic dishonesty due to overdependence on ChatGPT-generated content. 5) Absence of accountability mechanisms poses challenges when ChatGPT outputs inaccurate or biased information. Similarly, 191 (90.1%) mentioned a lack of such encouragement. This indicates that educators recognize the potential of AI in enhancing scholarly productivity and innovation. Only 21 (9.9%) of the respondents reported that their professors have recommended using ChatGPT or similar AI tools for academic or research-related tasks.

## Hypothesis of this study

**$H_0$:** There is no significant difference between research scholars' mean scores on UTAUT constructs and the neutral value, indicating no strong acceptance or rejection of ChatGPT for academic use.

**$H_1$:** There is a significant difference between research scholars' mean scores on UTAUT constructs and the neutral value, indicating a measurable perception (positive or negative) toward the adoption of ChatGPT for academic use.

Table 5 the descriptive statistics indicate that LIS research scholars exhibited positive perceptions across all three UTAUT constructs.

**Table 4.** Integrity and ethical consideration of using ChatGPT ($n = 212$).

| S.No. | Particular | No. of respondents (%) |
|---|---|---|
| | Do you have any institutional policy to use ChatGPT or any AI tool by students? | |
| 1. | Yes | 65 (30.7%) |
| 2. | No | 147 (69.3%) |
| | Have your professors recommended using ChatGPT or any AI tool for academic/research work? | |
| 1. | Yes | 21 (9.9%) |
| 2. | No | 191 (90.1%) |

**Table 5.** One-sample *t*-test results for UTAUT constructs among LIS research scholars ($n = 212$).Unified Theory of Acceptance and Use of Technology (UTAUT) analysis with a *t*-test.

| | N | Minimum | Maximum | Mean | Std. Dev | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|
| Variable | Statistics | Statistics | Statistics | Statistics | Statistics | Std. Error | Std. Error |
| PE | 212 | 31 | 3075 | 49.50 | 7.0 | 0.201 | 0.413 |
| EE | 212 | 17 | 1775 | 28.06 | 5.7 | 0.698 | 0.781 |
| SI | 212 | 17 | 1662 | 26.80 | 3.6 | 0.583 | 0.222 |
| Valid N (Listwise) | 212 | | | | | | |

| Variable | Mean ($\bar{X}$) | *T*-Statistic | *P*-Value | Interpretation |
|---|---|---|---|---|
| Performance Expectancy (PE) | 49.50 | 122.85 | <0.001 | Highly significant; ChatGPT is perceived as strongly performance-enhancing. |
| Effort Expectancy (EE) | 28.60 | 94.21 | <0.001 | Highly significant; users find it easy and effortless to use. |
| Social Influence (SI) | 26.80 | 114.10 | <0.001 | Highly significant; peer and academic community opinions positively influence usage. |

Performance Expectancy (PE) recorded the highest composite mean score ($M = 49.50$), suggesting that respondents perceived ChatGPT as a useful tool for enhancing research productivity, information retrieval, and academic performance. Effort Expectancy (EE) also demonstrated a high mean score ($M = 28.06$), indicating that respondents considered ChatGPT easy to learn and use for academic activities. Similarly, Social Influence (SI) showed a favorable mean score ($M = 26.80$), reflecting that peers, supervisors, and the academic environment positively influenced respondents' intention to adopt ChatGPT.

A one-sample *t*-test was conducted using the composite midpoint values of the respective UTAUT constructs to examine whether respondents' perceptions significantly differed from a neutral position. The results showed statistically significant positive perceptions for Performance Expectancy (PE), Effort Expectancy (EE), and Social Influence (SI) ($p < 0.001$). Therefore, the null hypothesis ($H_0$) was rejected in favor of the alternative hypothesis ($H_1$), indicating that LIS research scholars demonstrated significantly positive perceptions toward the adoption of ChatGPT for academic and research purposes.

- **PE** showed the highest **(Mean = 49.5)**, indicating that research scholars perceive a strong positive impact of ChatGPT on their academic performance.
- **EE (Mean = 28.6),** also demonstrated high significance, reflecting that scholars consider ChatGPT easy to use and user-friendly.
- **SI (Mean = 26.8),** suggests that peer influence and social acceptance play a vital role in motivating scholars to adopt ChatGPT in their research practices.

The result of null hypotheses stating that there is no significant perception of ChatGPT's usefulness, ease of use, or social influence are rejected. Therefore, it can be concluded that research scholars significantly perceive ChatGPT as useful, easy to use, and positively influenced by peer usage, affirming its acceptance and relevance in academic contexts.

## Recommendations

To propose recommendations and strategies for optimizing the integration of ChatGPT into academic and research practices among research scholars.

1. **Develop Comprehensive Institutional AI Policies:** The findings revealed that 69.3% of respondents reported the absence of institutional policies governing the academic use of ChatGPT and other AI tools. Universities should therefore formulate comprehensive institutional guidelines addressing acceptable AI use, authorship, citation practices, disclosure of AI assistance, plagiarism, privacy, and academic integrity. Clearly defined policies will reduce ambiguity and encourage the ethical use of generative AI in research.
2. **Strengthen AI Literacy and Research Skills among Research Scholars:** Although all respondents were aware of ChatGPT and a majority used it regularly for academic purposes, the study also found that many primarily relied on it for idea generation (80.7%), content generation (66.0%), and article summarization (61.3%). Universities should organize regular AI literacy programmes that train research scholars to use ChatGPT critically for literature searching, research design, academic writing, referencing, and responsible decision-making rather than merely content generation.
3. **Enhance Faculty Engagement in Responsible AI Adoption:** The study revealed that 90.1% of respondents had never received recommendations from their professors regarding the academic use of ChatGPT. This indicates limited faculty engagement in guiding responsible AI adoption. Universities should therefore conduct faculty development programmes that enable

teachers and research supervisors to guide scholars on appropriate AI use, ethical considerations, and discipline-specific applications of generative AI technologies.

4. **Promote Critical Evaluation of AI-Generated Information:** Since only 22.6% of respondents considered ChatGPT to be very accurate, while the majority perceived it as only somewhat accurate, research scholars should be encouraged to verify AI-generated information using peer-reviewed journals, books, institutional repositories, and other authentic scholarly resources before incorporating it into academic work. Universities and libraries should also provide training on evaluating the reliability and credibility of AI-generated content.
5. **Expand the Role of Academic Libraries in AI Support Services:** The results indicate that ChatGPT is increasingly being used for literature reviews, article summarization, and academic writing support. Academic libraries should therefore expand their services by introducing AI-assisted research support, awareness programmes, research consultations, and information literacy workshops. Librarians should guide users in combining AI tools with scholarly databases while emphasizing proper citation practices and ethical information use.
6. **Address Privacy, Data Security, and Ethical Concerns:** A considerable proportion of respondents expressed concerns regarding privacy and data security while using ChatGPT. Universities should establish clear guidelines discouraging researchers from uploading confidential research data, unpublished manuscripts, sensitive datasets, or personal information to public AI platforms. Training programmes should also emphasize responsible data management and compliance with research ethics.
7. **Improve Digital Research Infrastructure:** The study identified several functional limitations of ChatGPT, including the inability to provide reliable scholarly sources, limited access to real-time information, and inadequate support for graphical and image-based analysis. Universities should therefore complement AI tools with access to high-quality scholarly databases, institutional repositories, digital libraries, and subscription-based academic resources so that research scholars can validate AI-generated outputs using authentic evidence.
8. **Encourage Human–AI Collaboration Rather Than AI Dependence:** Although respondents demonstrated favorable perceptions toward Performance Expectancy, Effort Expectancy, and Social Influence, ChatGPT should be viewed as a supportive research assistant rather than a substitute for researchers' own analytical abilities. Universities should encourage scholars to use AI for brainstorming, summarization, language refinement, and information retrieval while ensuring that interpretation, critical analysis, originality, and final academic decisions remain the responsibility of the researcher.

## Conclusion

This study reveals a high level of awareness, primarily through informal sources like social media and peer interaction, but actual usage for academic work remains moderate. ChatGPT is mostly used for idea generation, drafting assignments, and exploring research topics. Using the UTAUT framework, the study observed positive perceptions in performance expectancy ($M = 49.50$), effort expectancy ($M = 28.06$), and social influence ($M = 26.80$), all of which were statistically significant ($p < 0.001$).

However, ethical concerns and institutional gaps were also evident 147 (69.3%) of respondents reported no institutional policy on AI use, despite 191 (90.1%) stating that faculty members recommended ChatGPT for research support. Despite its benefits such as 24/7 availability, time-saving assistance, and ease of access research scholars also reported key challenges, including the lack of credible literature sources 53 (85.5%), privacy concerns 43 (69.4%), and limitations in analyzing non-textual data like graphs and images 41 (66.1%). The study underscores the urgent need for structured policy frameworks, ethical guidelines, and AI literacy training in academic institutions to ensure responsible and effective use of ChatGPT in research. Future research should expand the sample across multiple universities and disciplines to enhance representativeness and build a comprehensive understanding of AI adoption in higher education.

Future research may adopt longitudinal research designs to examine changes in AI adoption behaviors over time, include participants from multiple academic disciplines and different categories of higher education institutions, and undertake cross-country comparative studies to explore cultural and institutional differences in the adoption of generative AI technologies. Future studies may also employ advanced analytical techniques such as Structural Equation Modeling (SEM) or other multivariate approaches to examine the causal relationships among technology adoption constructs.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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